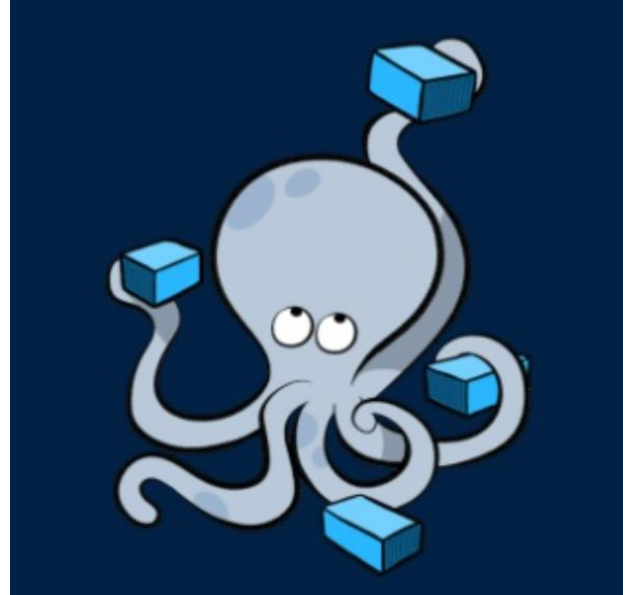


Docker Compose



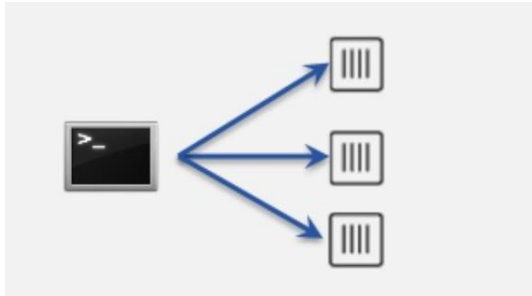
Agenda

- What is Docker Compose?
- 3 Steps of Docker Compose
- Building Blocks of Docker Compose
- Docker Compose File Format
- Networking in Docker Compose
- Environmental Variable under Docker Compose
- Installing Docker Compose

Docker Compose: Multi Container Applications

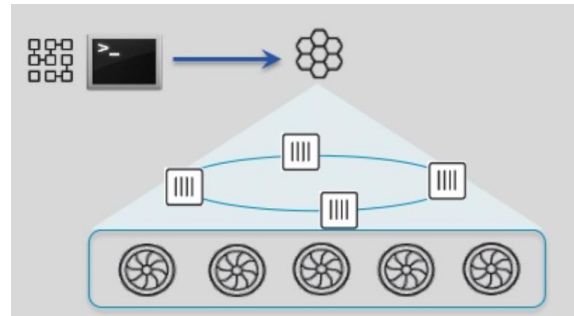
Without Compose

- Build and run one container at a time
- Manually connect containers together
- Must be careful with dependencies and start up order



With Compose

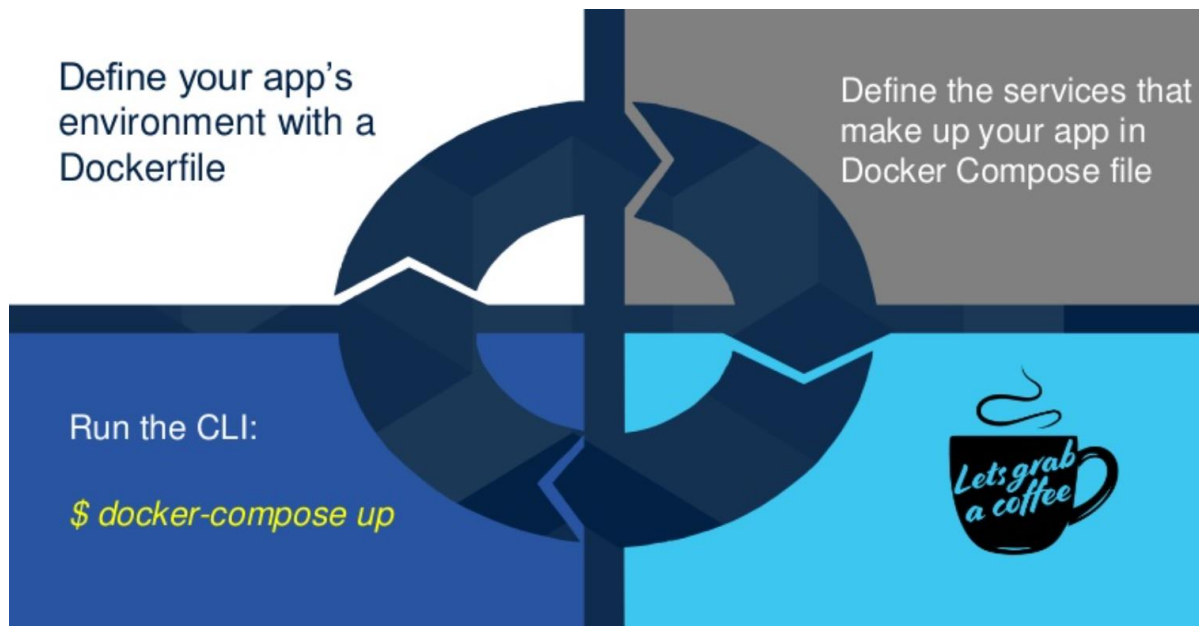
- Define multi container app in compose.yml file
- Single command to deploy entire app
- Handles container dependencies
- Works with Docker Swarm, Networking, Volumes, Universal Control Plane



What is Docker Compose?

1. A tool for defining and running multi- container Docker applications
2. With Compose, you use a YAML file to configure your application's services.
3. Compose works in all environments: production, staging, development, testing, as well as CI workflows.
4. With a single command, you create and start all the services from your configuration

Docker Compose is a 3 Steps



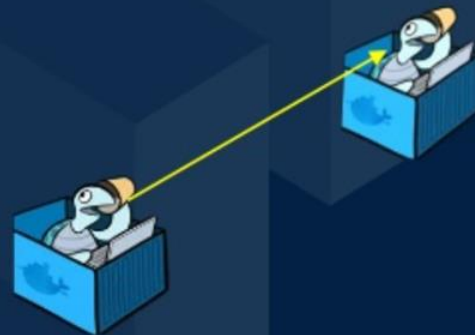
Building Blocks of Docker Compose



Services



Volumes



Networking

Docker Compose: Multi Container Applications



compose.yml

images

ports

volumes

links

containers:

web:

build: .

command: **python app.py**

ports:

- **"5000:5000"**

volumes:

- **../code**

environment:

- **PYTHONUNBUFFERED=1**

redis:

image: **redis:latest**

command: **redis-server --appendonly yes**

A Sample Docker Compose File

```
$ cat docker-compose.yml
version: '3'
services:
  web:
    build: .
    ports:
      - "5000:5000"
    volumes:
      - ./code
    environment:
      FLASK_ENV: development
  redis:
    image: "redis:alpine"
```


Docker Compose Compatibility Matrix

Compose file format	Docker Engine release
3.7	18.06.0+
3.6	18.02.0+
3.5	17.12.0+
3.4	17.09.0+
3.3	17.06.0+
3.2	17.04.0+
3.1	1.13.1+
3.0	1.13.0+
2.4	17.12.0+
2.3	17.06.0+
2.2	1.13.0+
2.1	1.12.0+
2.0	1.10.0+
1.0	1.9.1.+

Environmental Variable in Docker Compose: Using .env file

```
$ cat .env
TAG=v1.5

$ cat docker-compose.yml
version: '3'
services:
  web:
    image: "webapp:${TAG}"
```

Environmental Variable in Docker Compose: Using `docker-compose run` command

```
docker-compose run -e DEBUG=1 web python console.py
```

Networking in Compose

- A network called myapp_default is created.
- A container is created using web's configuration. It joins the network myapp_default under the name web.
- A container is created using db's configuration. It joins the network myapp_default under the name db.

```
version: "3"
services:
  web:
    build: .
    ports:
      - "8000:8000"
  db:
    image: postgres
    ports:
      - "8001:5432"
```

Compose for Swarm Mode

- The same Compose file can be used to deploy containerized apps on multi-host system
- The overlay driver creates a named network across multiple nodes in a swarm.
- The “global mode” enable at least one copy of apps running on every node of the swarm

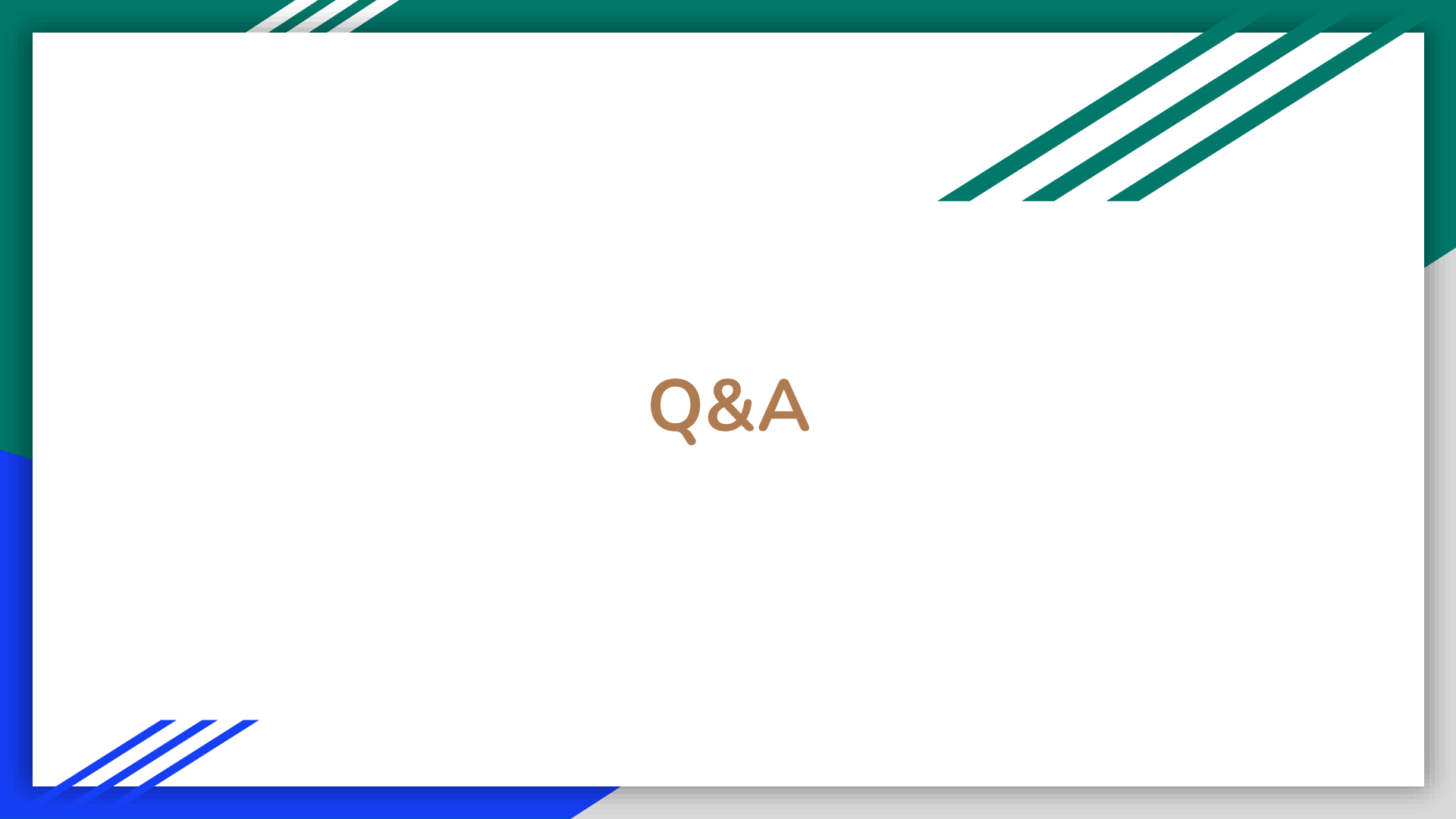
```
services:
  cadvisor:
    image: google/cadvisor:${CADVISOR_VERSION:-v0.24.1}
    networks:
      - monitoring
    volumes:
      - /var/run/docker.sock:/var/run/docker.sock,readonly
      - /:/rootfs
      - /var/run:/var/run
      - /sys:/sys
      - /var/lib/docker:/var/lib/docker
    deploy:
      mode: global
      resources:
        limits:
          cpus: '0.10'
          memory: 128M
        reservations:
          cpus: '0.10'
          memory: 64M
```

How to Install Docker Compose

```
$ curl -L https://github.com/docker/compose/releases/download/1.25.0/docker-compose-`uname -s`-`uname -m` -o /usr/local/bin/docker-compose
```

```
$ chmod +x /usr/local/bin/docker-compose
```

Note: Docker Desktop will automatically install the latest version of Docker Engine for you.



Q&A